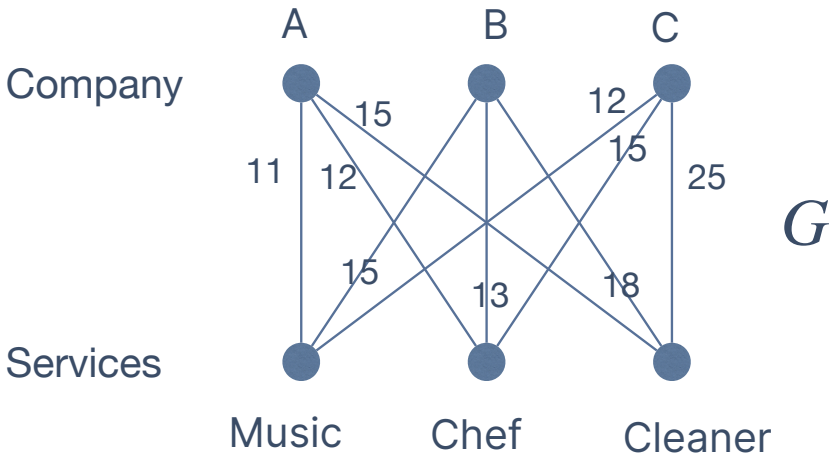
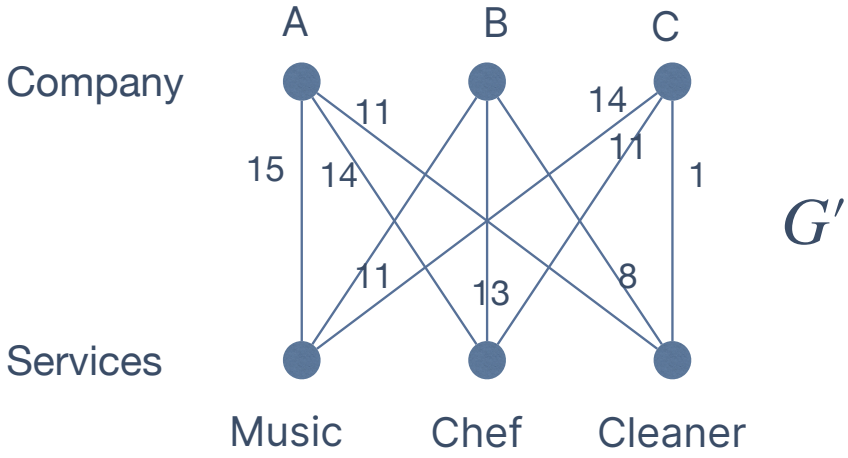
Company	Cost for Musician	Cost for Chef	Cost for Cleaner
A	11	12	15
B	15	13	18
C	12	15	25



Original Input graph G

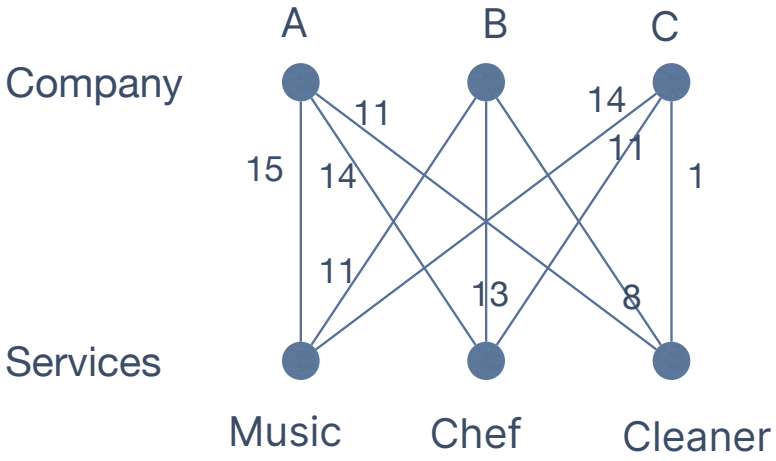
Solution

Company	Cost for Musician	Cost for Chef	Cost for Cleaner
A	15	14	11
B	11	13	8
C	14	11	1

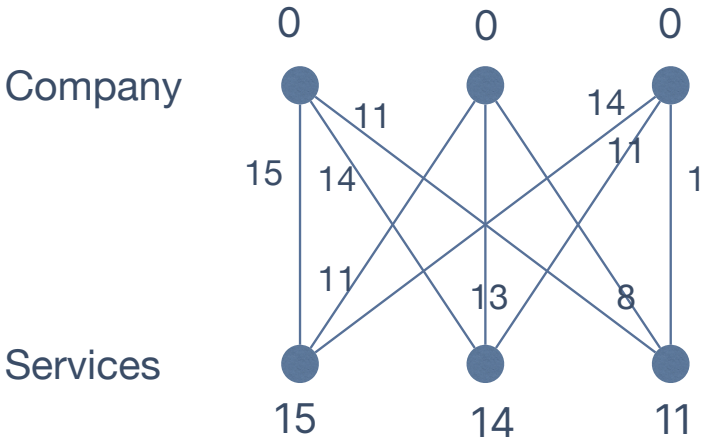


Input graph G' for max-weight matching

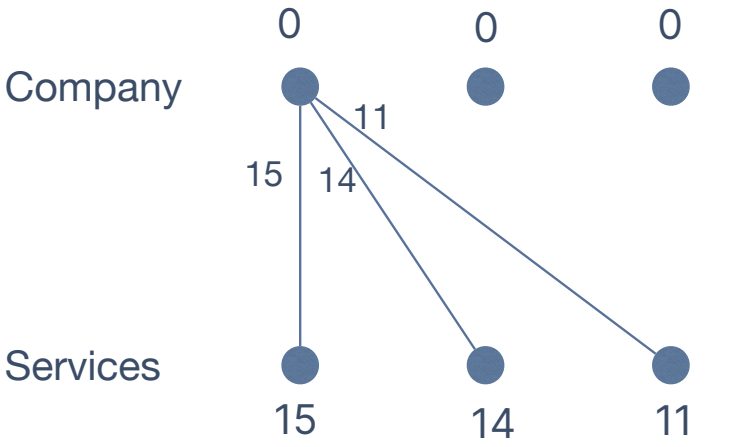
Solution



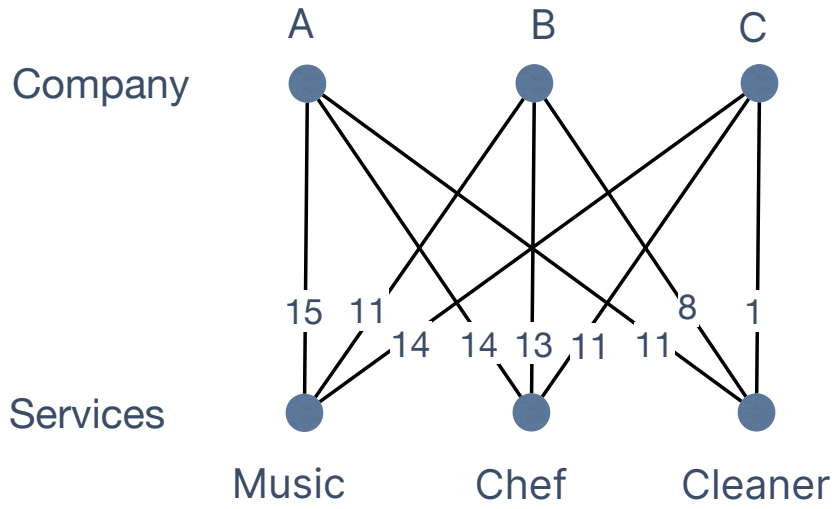
Feasible labeling



Equality subgraph G_l

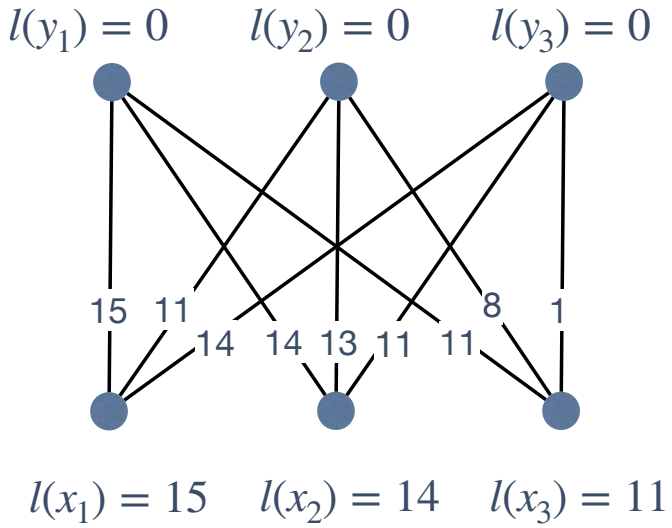


(a) Weighted complete bipartite graph of the problem

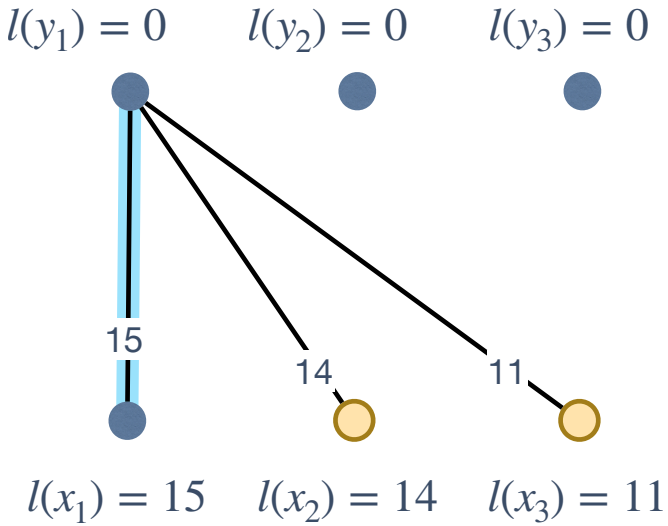


(b) Feasible Labeling and Equality Subgraph

G

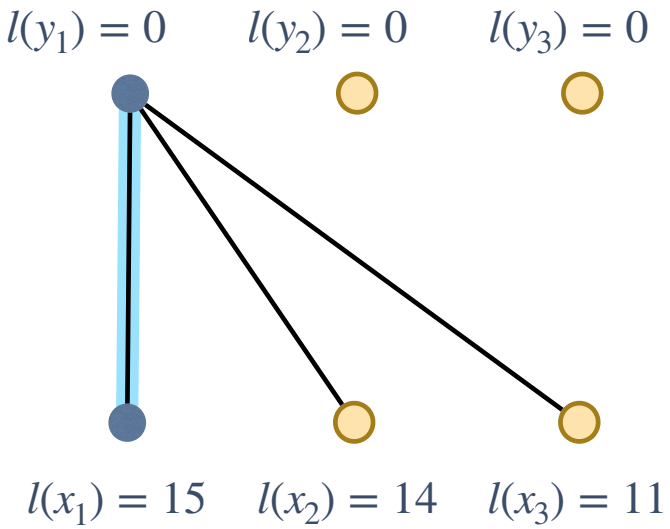


G_l and Initial matching M

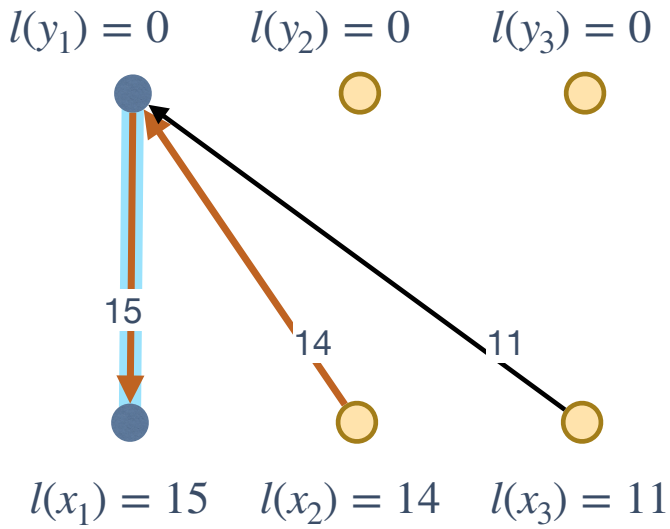


(c) BFS Forest on the Equality Subgraph

Equality Subgraph G_l

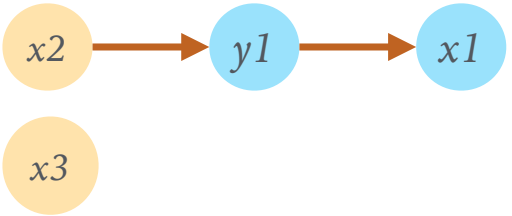


Digraph $G_{M,l}$



BFS forest

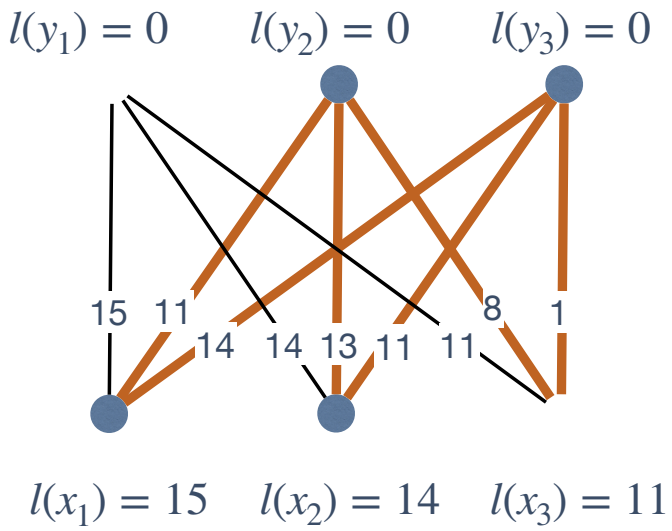
$F = (V_F, E_E),\ V_F = F_X \cup F_Y$



$F_X = \{x_1, x_2, x_3\}$

$F_Y = \{y_1\}$

(c) Updated Labels

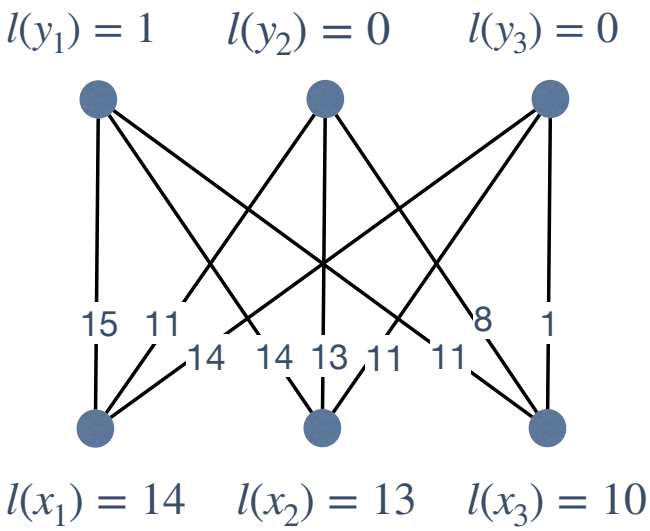


$$F_X = \{x_1, x_2, x_3\}$$
$$F_Y = \{y_1\}$$

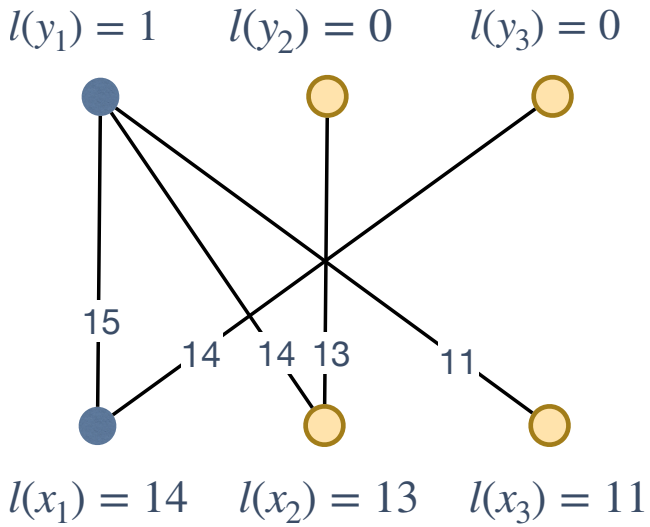
$$\delta = \min_{\substack{x \in F_X \\ y \in Y - F_Y}} l(x) + l(y) - w(x, y)$$
$$= \min\{15 - 11, 15 - 14, 14 - 13, 14 - 11, 11 - 8, 11 - 1\}$$
$$= 1$$

(c) Updated Equality Subgraph

$\delta = 1$

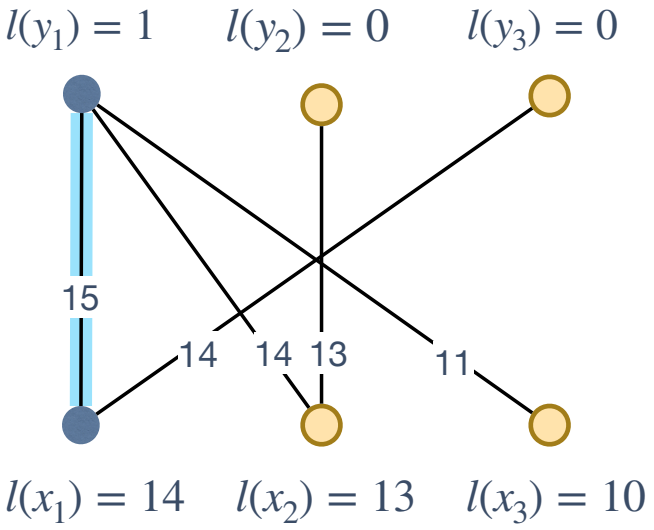


Updated G_l

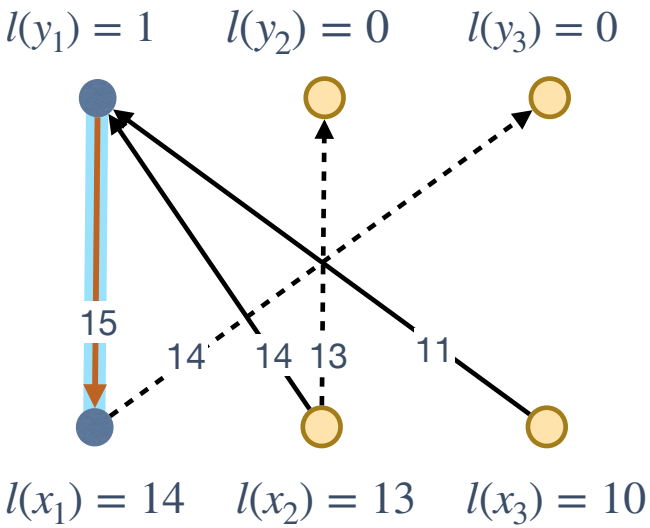


(c) BFS Forest on the Updated Equality Subgraph

Updated Equality Subgraph G_l



Digraph $G_{M,l}$



Two new edges added - (x1,y3), (x2,y2)
For (x1,y3), y3 not in FY and free, so an M-augmenting path is found.

BFS forest

$F = (V_F, E_F),\ V_F = F_X \cup F_Y$

$F_X = \{x_1, x_2, x_3\}$

$F_Y = \{y_1, y_3\}$

